ARCHITECTURE & IMPLEMENTATION GUIDE

AI Threat Intelligence Detection Chatbot

A lightweight RAG + OKF framework deployed in Google Colab for rapid security querying and threat analysis.

SentenceTransformers | Qwen2.5-0.5B Instruct | Gradio Interface

SYSTEM OVERVIEW

Three Core Pillars of the Architecture

1. OKF Knowledge Base

Structured Markdown storage with Frontmatter metadata (`type:`, `title:`).

2. Vector Retrieval Engine

Dense semantic embedding search powered by `all-MiniLM-L6-v2` cosine similarity.

3. Synthesis & UI Layer

Gradio dashboard driving local Qwen2.5 LLM generation with strict source bounds.

COMPONENT MAPPING

Technology Stack & Location Mapping

Tool / Library	Role in Pipeline	Where Used in Notebook
`google.colab.files`	File Upload Interface	`Cell 3` — Upload custom threat reports & notes
`SentenceTransformer`	Vector Embedding Search	`Cell 5` — Encodes query and threat chunks into vectors
`Qwen2.5-0.5B-Instruct`	Language Model Synthesis	`Cell 6` — Synthesizes short threat answers from retrieved notes
`Gradio (gr.Blocks)`	Interactive Web UI	`Cell 7` — Renders user chat interface with 4 example queries

PROCESS FLOW

Execution Lifecycle (Cell 1 to Cell 7)

1

Ingest & Upload

Cell 2 & Cell 3 create structure and accept custom files.

2

Parse & Validate

Cell 4 validates OKF schema and auto-updates index.md.

3

Vector Search

Cell 5 splits text by section headers and generates embeddings.

4

Agent Logic

Cell 6 filters low scores (<0.30) and prompts local LLM.

5

Gradio UI

Cell 7 renders 4 query examples and public link.

DATA ARCHITECTURE

OKF Knowledge Format & File Storage

```
# Example: knowledge/threats/ransomware_lockbit.md
---
type: ThreatIntel
title: LockBit Ransomware
---
# LockBit Ransomware
## Indicators of Compromise
Malicious PowerShell execution and hash records.
## Mitigation
Isolate affected endpoints immediately.
```

YAML Frontmatter Header

Stores metadata used by validation scripts and agent classification.

Section Split Boundaries (##)

Ensures vector search chunks text by distinct tactical ideas rather than random tokens.

GUARDRAILS & PRECISION

Strict Hallucination Prevention Strategy

0.30

Minimum Cosine Similarity Threshold

Queries with similarity scores below 0.30 immediately trigger fallback defenses:

"I could not find matching threat intelligence in my database."

USER INTERFACE

Gradio Interface & Included Example Prompts

EXAMPLE QUERY 1

"How do I mitigate LockBit ransomware?"

EXAMPLE QUERY 2

"What are the affected systems for CVE-2026-XYZ?"

EXAMPLE QUERY 3

"What are the indicators of compromise for LockBit?"

EXAMPLE QUERY 4

"How do we patch CVE-2026-XYZ?"

Ready to Deploy in Google Colab

Run Cells 1 to 7 sequentially to launch your custom AI threat detection assistant with custom upload capabilities.

RAG Engine • OKF Format • Gradio Dashboard